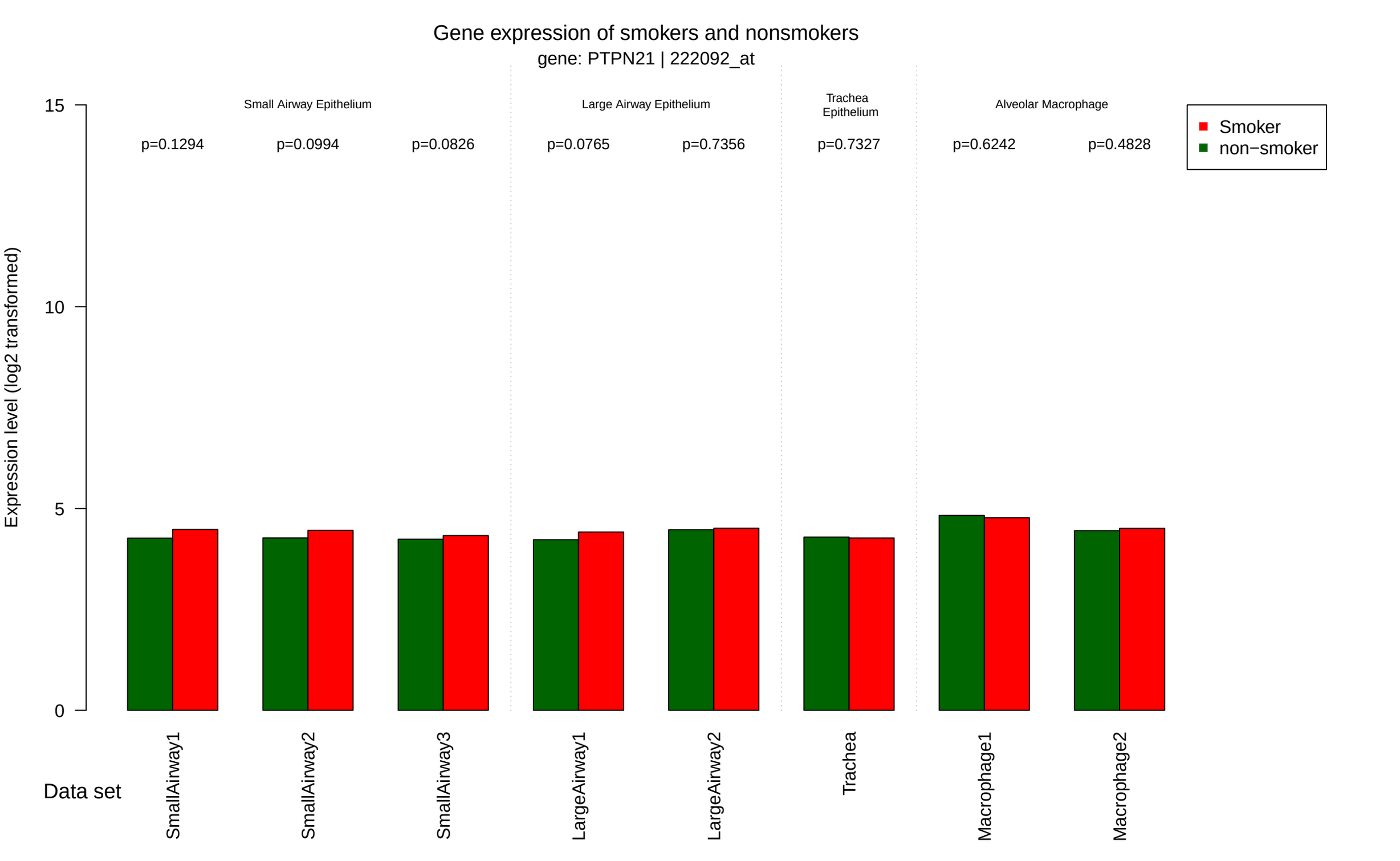
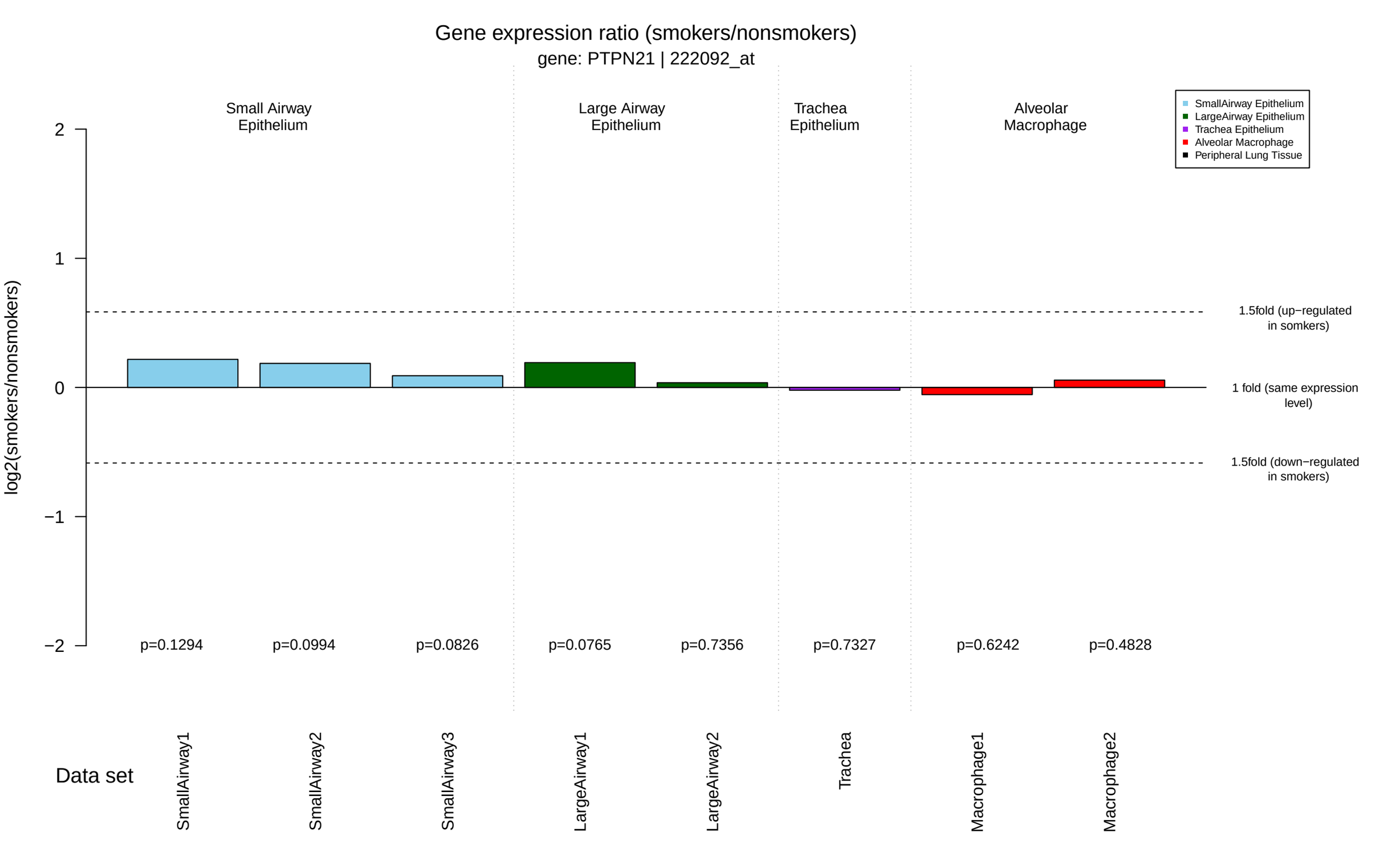
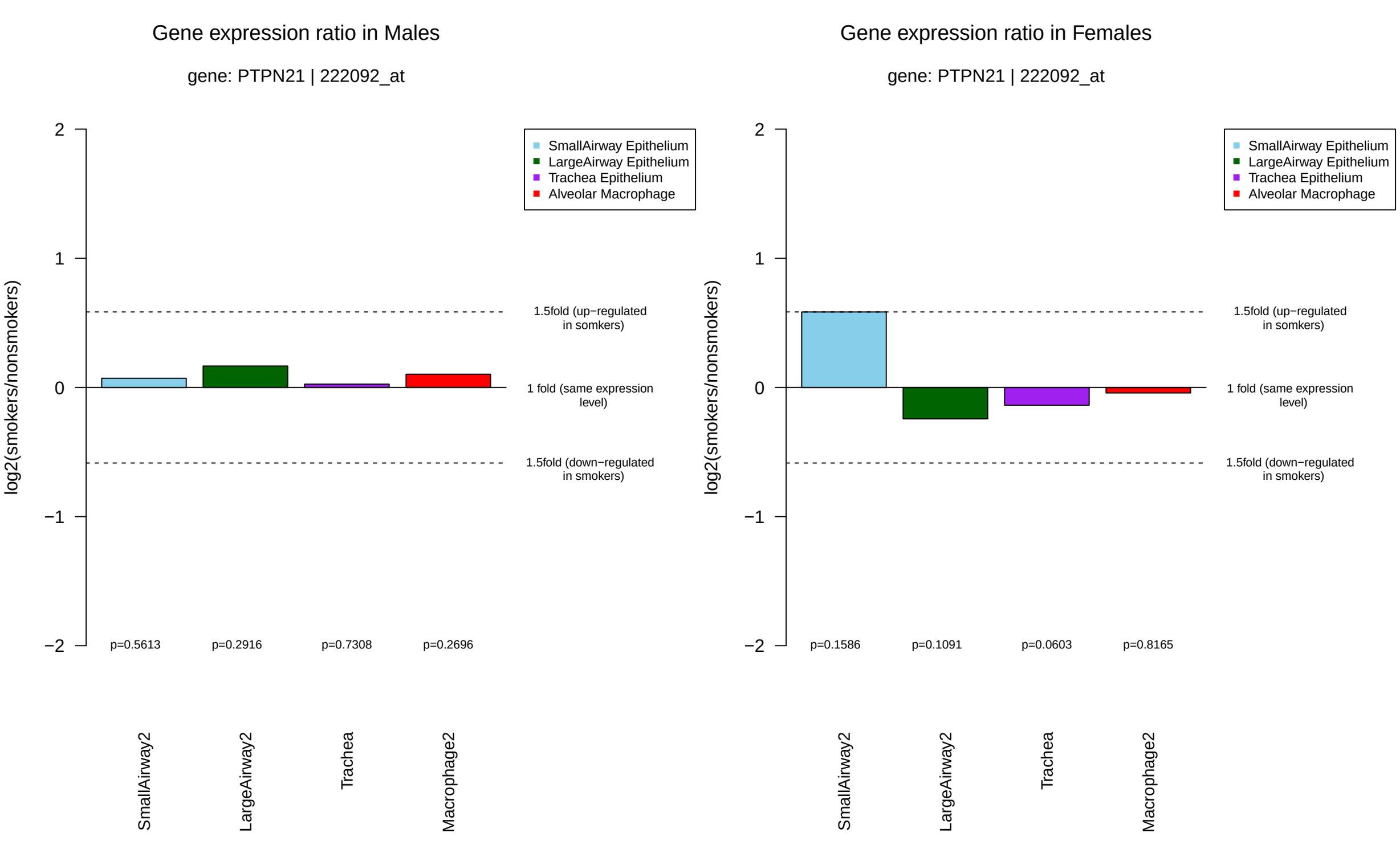
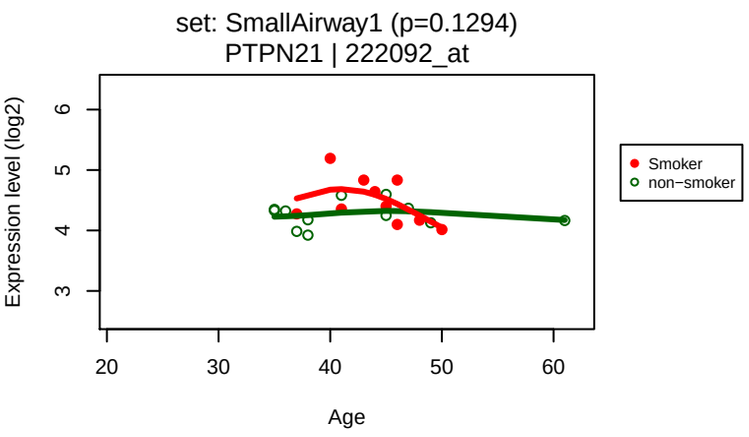
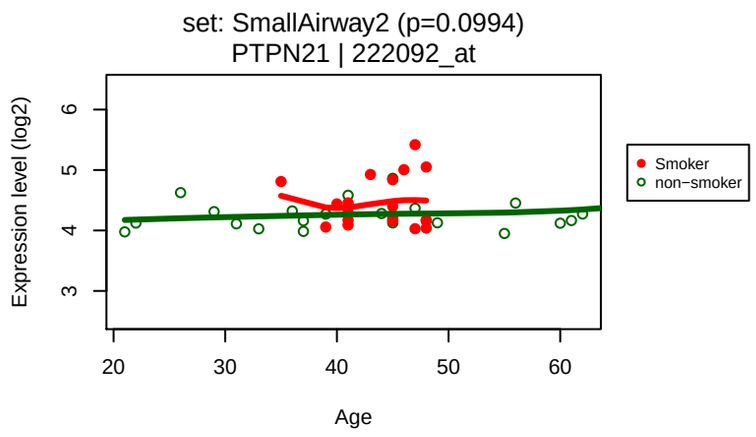
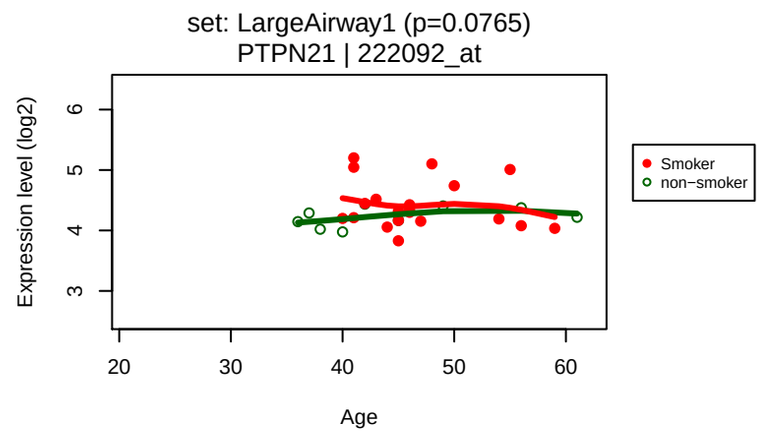

Gene expression of smokers and nonsmokers
gene: PTPN21 | 222092_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1294
p=0.0994
p=0.0826
p=0.0765
p=0.7356
p=0.7327
p=0.6242
p=0.4828
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

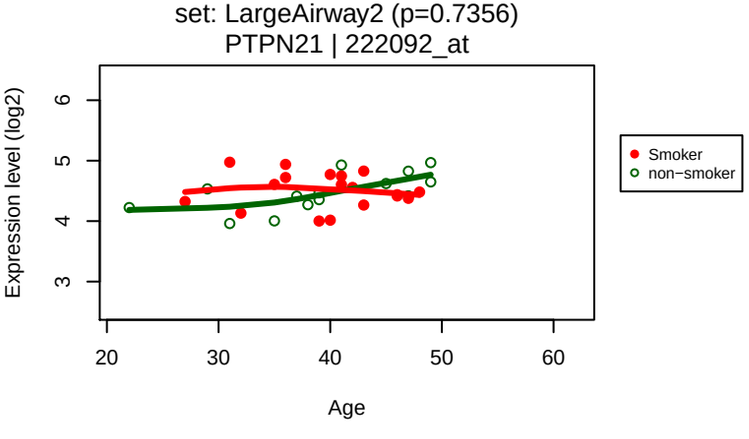

Gene expression ratio (smokers/nonsmokers)
gene: PTPN21 | 222092_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1294
p=0.0994
p=0.0826
p=0.0765
p=0.7356
p=0.7327
p=0.6242
p=0.4828
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

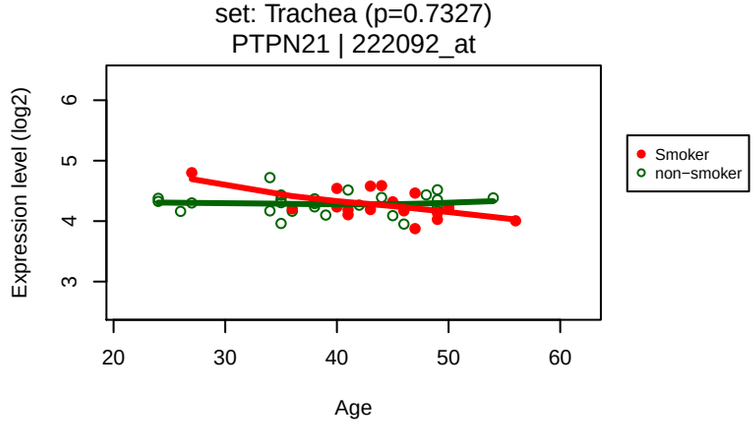

Gene expression ratio in Males
gene: PTPN21 | 222092_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5613
p=0.2916
p=0.7308
p=0.2696
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN21 | 222092_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1586
p=0.1091
p=0.0603
p=0.8165
SmallAirway2
LargeAirway2
Trachea
Macrophage2

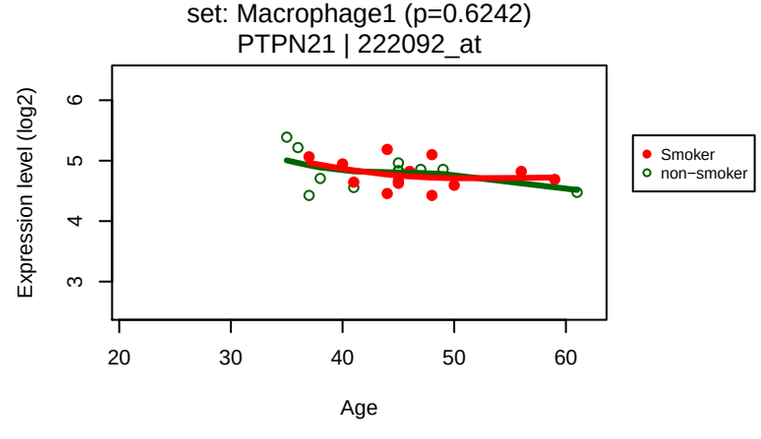

set: SmallAirway1 (p=0.1294)
PTPN21 | 222092_at
Expression level (log2)
Age
Smoker
non−smoker

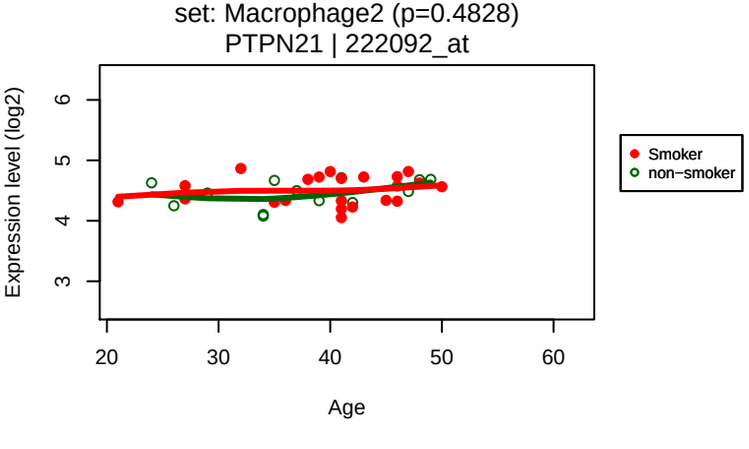

set: SmallAirway2 (p=0.0994)
PTPN21 | 222092_at
Expression level (log2)
Age
Smoker
non−smoker

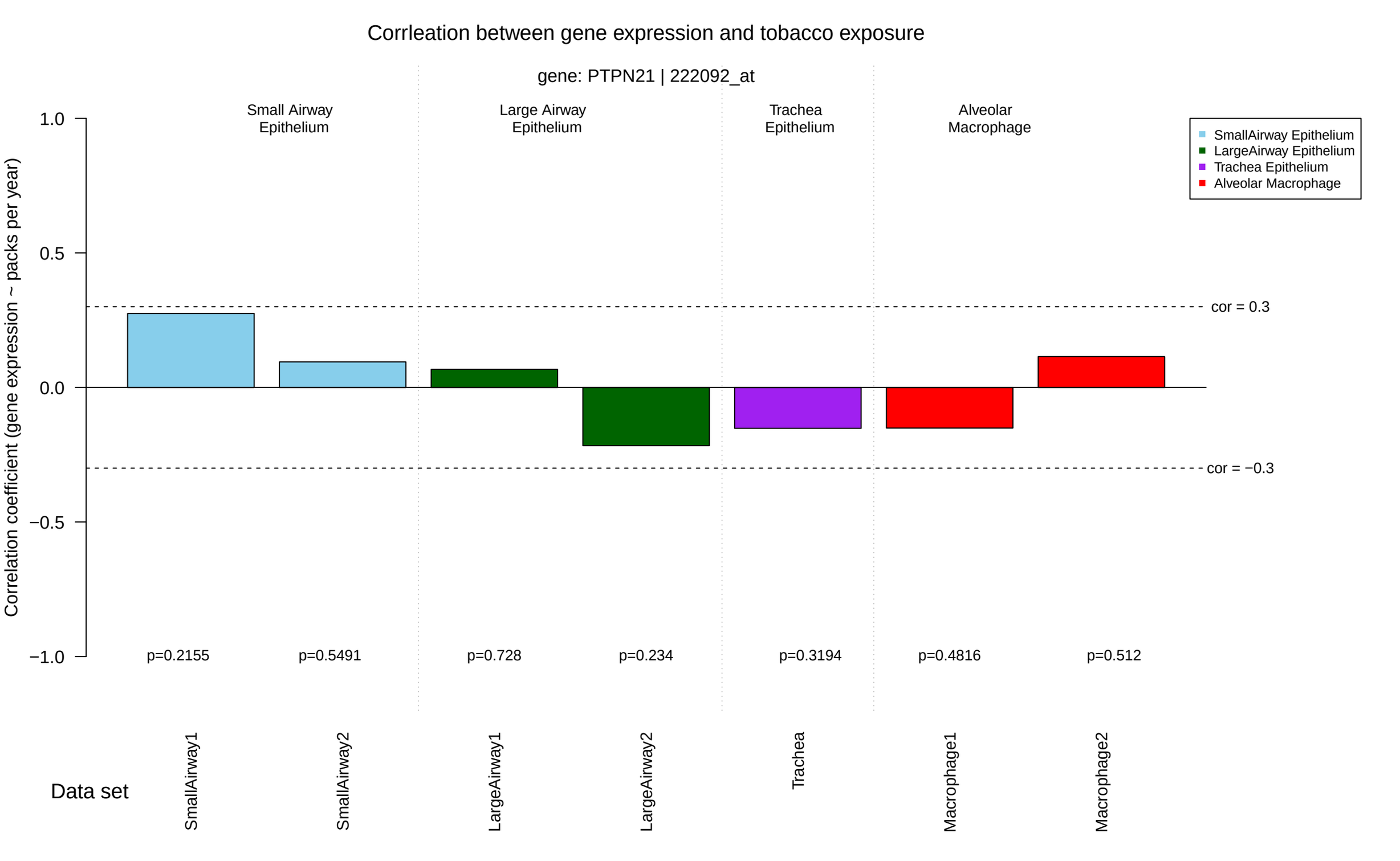

set: LargeAirway1 (p=0.0765)
PTPN21 | 222092_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7356)
PTPN21 | 222092_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7327)
PTPN21 | 222092_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6242)
PTPN21 | 222092_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4828)
PTPN21 | 222092_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN21 | 222092_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2155
p=0.5491
p=0.728
p=0.234
p=0.3194
p=0.4816
p=0.512
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2